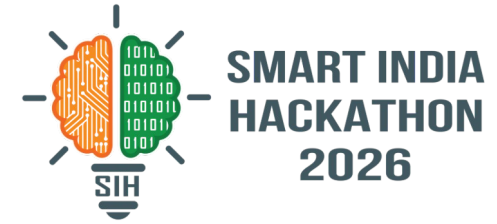


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – 26166
- **Problem Statement Title** – Sub-pixel correspondence between Chandrayaan-2 optical images and lunar reference images, robust to illumination, viewpoint and scale
- **Theme** – Space Technology
- **PS Category** – Software
- **Team ID** – 800C4B
- **Team Name** – Bugs Janta Party



SETU

Geometry for scale and viewpoint. Physics for the Sun. Learning only for what is left.



SETU: registration that spends the geometry and removes the Sun



PROPOSED SOLUTION

PS 26166 names three variations. Two of them are not appearance problems.

- Scale and viewpoint are geometry, and Chandrayaan-2 ships the geometry in every product. SETU ortho-projects both images onto one map projection at one ground sampling distance on SLDEM2015, which collapses ratios up to 160x and tens of degrees of viewpoint into a residual planar shift.
- Illumination is real appearance, so we delete it. The reference is re-rendered from the terrain model under the source image's own solar azimuth, elevation and emission angle, using Lunar-Lambert reflectance with McEwen limb darkening and ray-cast shadows. Matching then happens between two images that agree about where the Sun is.
- Whatever survives goes to two independent matchers. A pretrained cross-modality network and a phase-congruency structural matcher run separately. A correspondence is kept only when both agree within 2 px, or when one has a provably sharp correlation peak.

WHY THIS ADDRESSES THE PROBLEM

- Chandrayaan-2 geolocation is out by kilometres (published OHRC reprocessing puts SPICE positions about 4 km from truth), while the requirement is sub-pixel. Geometry closes the kilometres. Physics closes the illumination. Learning is left only with the residual.
- Uniformity is an objective, not a by-product. Points are quota'd over a lattice on the true overlap, spread inside each cell by farthest-point selection, and empty cells are re-searched against the model once it is known.
- Every tie point carries a 2x2 covariance that weights the robust fit, sets the outlier threshold, and is written into the output file. That is what makes a match list ingestible.

WHAT MAKES IT HARD, MEASURED

Same crater field, same viewpoint, Sun moved from east to west. Correlation between the two images: **-0.936**.

Not weakly correlated. Inverted. Every gradient a descriptor keys on has changed sign, which is why SIFT, ORB and a MegaDepth-trained network fail here rather than degrade.

WHAT SETU DOES ABOUT IT

Re-illuminate the reference and that same pair correlates at **+0.970**.

Tie-point RMSE against exact truth: **0.361 px**

SIFT, on the same pairs: **150.1 px**

LoFTR, on the same pairs: **10.25 px**

Correspondences within 3 px of truth: **100.0%**

Inlier ratio after MAGSAC++: **100.0%**

Lattice coverage at 150 points: **0.81**

LoFTR coverage at the same count: **0.40**

Measured over 31 benchmark pairs whose true geometric relationship is known exactly, not fitted.

TECHNOLOGIES

Language and arrays: Python 3.11, NumPy, SciPy

Raster and geodesy: rasterio, GDAL, pyproj, shapely, lunar CRS on R = 1,737,400 m

Planetary I/O: pds4_tools, pvl, spiceypy, PDS4 and PDS3 readers, windowed access

Classical vision: OpenCV with USAC_MAGSAC, scikit-image

Structural features: vendored Kovesi phase congruency, MIM, CFOG, LNIIFT in NumPy and FFT

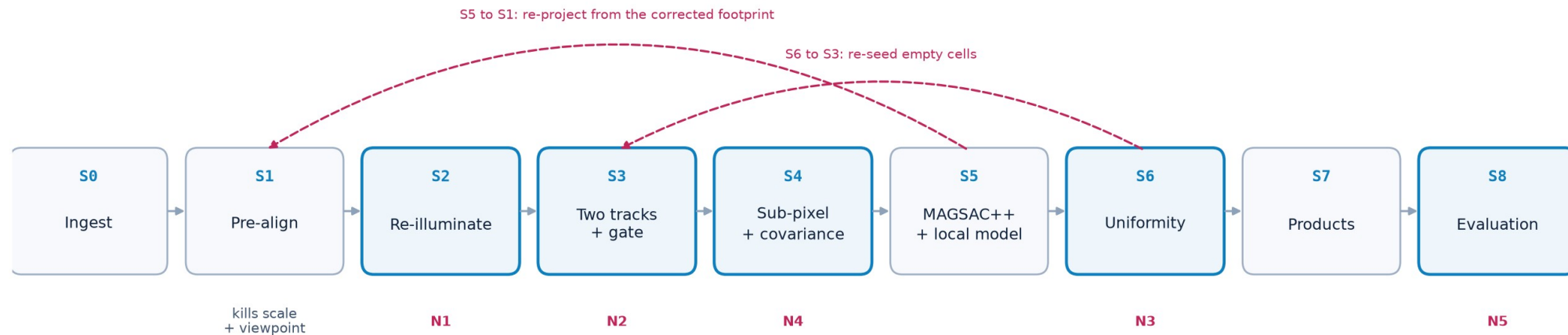
Deep matching: PyTorch, kornia; MatchAnything, LoFTR, LightGlue, XFeat behind one adapter

Service and demo: FastAPI, uvicorn, React, Vite, TypeScript, GSAP

Packaging: Docker image, GitHub Actions running lint, tests and an end-to-end pair

METHODOLOGY

- **Spend the geometry first.** Choose the working CRS from the footprint, work at the coarser of the two GSDs, and never upsample the coarser image to meet the finer one. Off-nadir imagery gets an explicit terrain-parallax correction, which at 25 degrees and 200 m of relief is 373 OHRC pixels.
- **Then remove the Sun.** Render the reference at the source's solar geometry. Shadows come from a horizon sweep along the sun azimuth, which is $O(N)$ per azimuth and needs no mesh library.
- **Then match, twice, and gate.** Track A is a pretrained cross-modality network on the harmonised imagery. Track B is phase-congruency detection with maximum-index-map description and CFOG template refinement, on the structural maps.
- **Then measure everything.** The harness was built before the matcher, so no tuning decision was made against intuition.



Two feedback edges are implemented, not described. Once a global transform exists, pre-alignment re-runs from the corrected footprint. Cells that fail their quota are re-searched at a lowered threshold inside a 5 px window, which is what actually delivers uniformity: coverage rises from 0.33 to 0.90 on a representative run.

FEASIBILITY: IT IS BUILT AND IT IS MEASURED

- All nine stages implemented, both feedback edges included, running end to end on CPU.
- 31 benchmark pairs, 12 methods, every metric with a bootstrap confidence interval over pairs.
- 40 tests, including photometric limiting cases, Clark-Evans calibrated against a Poisson process, and a full pipeline run checked against a known transform.
- A registration takes about 5.6 s per pair on a laptop, with no GPU required.
- Docker image plus CI that builds it and waits for the container's health check.

THE PROPERTY WE ARE PROUDEST OF: IT DECLINES RATHER THAN LIES

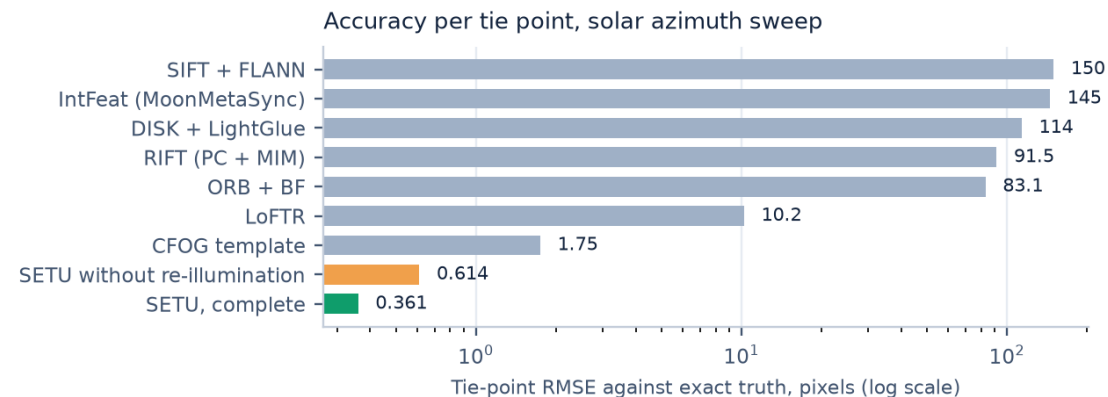
At an eightfold scale ratio the deep matcher returns 144 confident correspondences with **0.0% precision**, a median error of 92 px. The agreement gate rejected every one of them and SETU returned no registration at all. For a system whose output might inform a landing site, refusing to answer is the correct behaviour and a plausible wrong answer is the dangerous one. This is why the two tracks are independent and why the gate exists.

VIABILITY

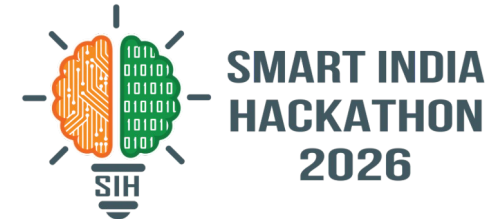
- Nothing hard-codes a payload. OHRC, TMC-2, IIRS, NAC, Kaguya TC and WAC all enter through one Product abstraction.
- The reference policy is an inspectable table, so IIRS is matched against Kaguya TC or a hillshade rather than being forced against 0.5 m NAC.
- Illumination is never guessed. If it cannot be resolved from a backplane, from SPICE, or from label keywords, the run stops.

CHALLENGES AND WHAT WE DID ABOUT THEM

- **PRADAN needs an account and the OHRC archive is over 200 GB.** P0 to P2 were built entirely against the controlled benchmark, so no phase was ever blocked on a download. The PDS4, PDS3 and GeoTIFF readers are written and tested against real label structures.
- **SPICE and ISIS integration can consume days.** Tier B, a corner fit with terrain parallax, is the default and is sufficient. Tier A is behind a capability check that degrades and says so.
- **A 59 m DEM cannot render 25 cm OHRC photorealistically.** It is not asked to. The render constrains global and mid-frequency alignment; sub-pixel accuracy comes from S4 on the real image pair. We state this rather than implying the render is photoreal.
- **Repetitive crater fields make deep matchers confidently wrong.** That is what the gate is for, and it is measured below.



IMPACT AND BENEFITS



IMPACT ON THE TARGET USER

- **Archived Chandrayaan-2 data becomes geometrically usable.** Kilometre-level geolocation is the reason OHRC, TMC-2 and IIRS products are hard to combine with each other or with LRO. Correcting that turns an archive of individually good images into a co-registered one.
- **A tie-point list an agency can actually ingest.** Every point carries its own covariance, so a downstream bundle adjustment or DEM pipeline can weight it properly instead of treating all matches as equal.
- **A measurement of spacecraft pointing, for free.** The per-row jitter spline fitted for pushbroom payloads reports the amplitude and dominant period of attitude jitter, which falls out of registration as a by-product.
- **Site selection and change detection.** Uniform, uncertainty-tagged correspondence over the whole overlap is what landing site characterisation and repeat-pass comparison need.

BENEFITS

- **Scientific.** The method generalises past the Moon. Any airless body with a shape model and seleno-tagged imagery has the same illumination problem, so Mars, Mercury and asteroid campaigns inherit the approach unchanged.
- **Operational.** Runs on a laptop with no GPU. A registration finishes in seconds, and the whole install is one Docker image, which is the part most likely to fail on the day.
- **Economic.** Nothing proprietary. Every dependency is open source and every reference dataset is public, so there is no licence to renew and no vendor to depend on.
- **Methodological.** The benchmark generator ships with the system, so a future team can measure against exact truth rather than against a screenshot.

TRY IT

Live demo: <https://arya-patil686.github.io/setu-sih2026/>

Three complete registrations at increasing difficulty, the pipeline played stage by stage with the re-illumination as a visible step, tie points coloured by residual, and the full leaderboard with confidence intervals.

HEADLINE NUMBERS

0.361 px

tie-point RMSE vs exact truth

100%

matches within 3 px of truth

100%

inlier ratio after MAGSAC++

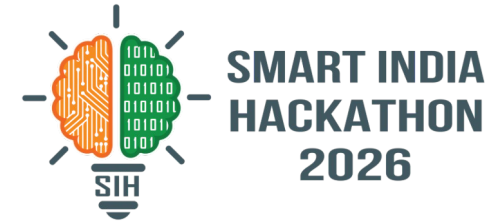
0.81

coverage at 150 points

416x

better than SIFT per tie point

RESEARCH AND REFERENCES



RESEARCH THIS BUILDS ON

- **He et al.** MatchAnything: Universal Cross-Modality Image Matching with Large-Scale Pre-Training. TPAMI 2026, arXiv:2501.07556. Track A weights.
- **Li, Hu and Ai.** RIFT: Multi-Modal Image Matching Based on Radiation-Variation Insensitive Feature Transform. IEEE TIP 29:3296, 2020. Track B descriptor.
- **Ye et al.** Structural similarity for multimodal remote sensing (HOPC), IEEE TGRS 55(5) 2017, and CFOG, ISPRS 2019.
- **Kovesi.** Image Features from Phase Congruency, 1999. Vendored, because phasepack breaks on modern NumPy.
- **Kumar, Kaushal and Murthy.** MoonMetaSync: Lunar Image Registration Analysis, arXiv:2410.11118, 2024. The lunar-specific floor we set out to beat.
- **Tungathurthi.** Geodetically Anchored 0.30 m DEM of the Chandrayaan-3 Vikram Landing Site, arXiv:2602.14993, 2026. Documents the 4 to 6 km OHRC geolocation error.
- **Barker et al.** SLDEM2015, a new lunar DEM from LOLA and SELENE Terrain Camera. Icarus 273:346, 2016. The shape model.
- **Chowdhury et al.** Chandrayaan-2 Orbiter High Resolution Camera. Current Science 118(4):560, 2020, with companion papers on TMC-2 and IIRS.
- **Verma, Chauhan and Chauhan.** Lunar surface temperature and thermal emission correction from Chandrayaan-2 IIRS. Icarus 383:115075, 2022.
- **Barath et al.** MAGSAC++, CVPR 2020. The robust estimator, with an adaptive threshold derived from our own per-point covariances.

DATA SOURCES

- **Chandrayaan-2 L1:** OHRC, TMC-2 and IIRS with SPICE kernels, via PRADAN (pradan.issdc.gov.in/ch2)
- **LRO NAC and NAC DTMs:** via LROC and the ODE search (ode.rsl.wustl.edu/moon)
- **SLDEM2015:** 512 ppd, about 59 m/px, from imbrum.mit.edu
- **Kaguya TC ortho and DEM:** JAXA DARTS, the resolution-appropriate reference for IIRS
- **LROC WAC:** global mosaic at about 100 m

THE WORK ITSELF

Live demo: <https://arya-patil686.github.io/setu-sih2026/>

Source: <https://github.com/Arya-Patil686/setu-sih2026>

Reproducing every number on these slides takes two commands:

```
python experiments/run_sweeps.py
```

```
python scripts/build_demo_bundle.py
```

The benchmark's ground truth is exact by construction: both images of every pair are rendered from one terrain model under a transform that is known rather than estimated. Results here are on that benchmark. The PDS4 and PDS3 readers, the SPICE path and the sensor models are implemented and waiting on archive access, not on code.